PORTFOLIO PROJECT · TEXT MINING · PUBLISHED ON SSRN

Can the words predict the ruling?

Predicting motion outcomes in 1,535 federal court opinions with SAS text mining and a decision tree

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D. Conn. opinions · 1999–2006

THE DATA

1,535 court opinions, collected and labeled from scratch

Step 1

Scrape

A Python scraper pulled every opinion PDF from the U.S. District Court for Connecticut, 1999 to 2006.

Step 2

Extract

PyPDF2 turned each PDF into text. Case number, judge and date came from the headers.

Step 3

Label

Keyword rules tagged each case: motion outcome, expert witness, §1983 claim, damages.

Civil cases: housing disputes, §1983 civil-rights claims, summary-judgment motions and administrative appeals.

TEXT MINING IN SAS ENTERPRISE MINER

From legal prose to 22 model-ready features

Parse

Keep nouns, verbs and adjectives. Stem words, merge synonyms.

→

Clean

Custom stop list for scan debris like "tion" and "fo".

→

Filter

Drop terms in fewer than 5 opinions. Entropy weighting.

→

Cluster

Group co-occurring terms. SVD yields 26 components.

→

Prune

Drop 4 garbled components. Keep 22.

Each component is a weighted mix of terms that tend to appear together, so each one behaves like a topic.

THE MODEL

A decision tree, because a lawyer can follow it

Target

Accepted or rejected, grouped from seven raw outcomes such as granted, denied and dismissed.

Inputs

22 text components plus case facts: expert witness, §1983 citation, judge.

Split

80% train, 15% test, 5% validation. Moved from 60/20/20 when early runs had too little to learn from.

RESULTS

About 7 in 10 unseen motions classified correctly

70.3%

accuracy on the test set, cases the model never saw

77.5%

accuracy on the validation set

79.7%

accuracy on the training set

WHAT THE MODEL LEARNED

The text does the heavy lifting

Text splits first

Two text components, SVD8 and SVD4, form the top of the tree. Case facts only refine the lower branches.

Shallow is enough

Lift is concentrated near the root. Deeper splits add little, so a pruned tree is safer.

Some overfitting

Accuracy drops from 79.7% in training to 70.3% on unseen cases.

NEXT STEPS

Making the result harder to fool

Hide the answer

Labels came from keywords in the same text. Strip those phrases so the tree can't relearn the rules.

Report a baseline

Set 70.3% against simply guessing the more common outcome every time.

Validate properly

5% validation is only about 77 cases. Cross-validation gives steadier estimates.

Published with Amanda Herbert and Emmanuel Chigowe: "Text Analysis in the Legal Domain: Uncovering Patterns in Court Rulings and Legal Opinions Using AI", SSRN, 2025.